3-6-26 (WEEK 2)

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We consider a field K, s.t. $K/\mathbb{Q}$: number field (fin ext of $\mathbb{Q}$), $\mathcal{O}_K$: free $\mathbb{Z}$-module of rank $[K:\mathbb{Q}]$

THEOREM

For K as above, we have:

(i) $\forall$ nonzero ideal $\mathfrak{a} \trianglelefteq \mathcal{O}_K$, $\mathcal{O}_K/\mathfrak{a}$: finite

(ii) $\forall$ nonzero prime ideal $\mathfrak{p} \trianglelefteq \mathcal{O}_K$, $\mathfrak{p}$: maximal

(iii) $\forall$ nonzero prime ideal $\mathfrak{p} \trianglelefteq \mathcal{O}_K$, $\mathfrak{p} \cap \mathbb{Z} = \langle q \rangle$ for some prime number q

Proof

(i) By structure thm, $\exists$ a $\mathbb{Z}$ basis $\{e_i\}$ of $\mathcal{O}_K$ s.t. $\mathfrak{a} = \bigoplus_{i=0}^{n} \mathbb{Z} b_i e_i \subseteq \bigoplus_{i=0}^{n} \mathbb{Z} e_i = \mathcal{O}_K$, $b_i \in \mathbb{Z}_{\geq 0}$

$\forall 0 \neq \alpha \in \mathfrak{a}$, we have $\bigoplus_{i=1}^{n} \mathbb{Z}\alpha e_i \subseteq \mathfrak{a} \subseteq \bigoplus_{i=1}^{n} \mathbb{Z} e_i \Rightarrow \mathrm{rk}_{\mathbb{Z}} \mathfrak{a} = n$. As $b_i \in \mathbb{Z}_{\geq 0}$, thus $\mathcal{O}_K/\mathfrak{a} \cong \bigoplus_{i=1}^{n} \mathbb{Z}/b_i\mathbb{Z}$

(ii) By (i), $\mathcal{O}_K/\mathfrak{p}$: finite domain $\Rightarrow$ field

(For $x \neq 0 \in \mathcal{O}_K/\mathfrak{p}$, $m_x : \mathcal{O}_K/\mathfrak{p} \to \mathcal{O}_K/\mathfrak{p}$ is inj since $\mathcal{O}_K/\mathfrak{p}$: domain. As it's finite, m_x is onto, hence isom)

(iii) Here, $\mathfrak{p} \cap \mathbb{Z}$: maximal $\Leftrightarrow$ $\mathfrak{p}$: maximal. By (ii), $\mathfrak{p}$: maximal. $\therefore \exists$ prime $q \in \mathbb{Z}$, s.t. $\mathfrak{p} \cap \mathbb{Z} = \langle q \rangle$ $\square$

CONCLUSION

(i) $\mathcal{O}_K$ is noetherian (Each submodule of $\mathcal{O}_K$ is finitely generated (ACC on submodules))

(ii) $\mathcal{O}_K$ is integrally closed ($\mathrm{Frac}(\mathcal{O}_K) = K$, $K \ni x$ integral over $\mathcal{O}_K \Rightarrow x$ integral over $\mathbb{Z} \Rightarrow x \in \mathcal{O}_K$) (by prev lemma)

(iii) All nonzero prime ideals in $\mathcal{O}_K$ are maximal ($\langle 0 \rangle \subsetneq \mathfrak{p} \Rightarrow \dim \mathcal{O}_K = 1$)

DEFINITION

An integral domain ($ab=0 \Rightarrow a=0$ or $b=0$) satisfying (i), (ii), (iii) as above is a Dedekind domain

QUESTION

Say p: prime number, $\mathbb{Z} \subseteq \mathcal{O}_K$. Which max ideal $\mathcal{O}_K$ contains $p\mathcal{O}_K$?

EXAMPLE

For $d \equiv 2, 3 \pmod 4$, $K = \mathbb{Q}(\sqrt{d})$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}] \cong \mathbb{Z}[x]/(x^2-d)$

If $\mathfrak{a} \trianglelefteq \mathcal{O}_K$, s.t. $\mathfrak{a} \cap \mathbb{Z} = (p)$, then we have:

$$\begin{tikzcd} \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \arrow[r, hook] \arrow[d] & \mathcal{O}_K/\mathfrak{a} \\ \mathcal{O}_K/p\mathcal{O}_K = \mathbb{F}_p[x]/(x^2-d) & \end{tikzcd}$$

Then,

$$\mathcal{O}_K/p\mathcal{O}_K = \begin{cases} \mathbb{F}_p \times \mathbb{F}_p & \text{if } d \in (\mathbb{F}_p^\times)^2 \text{ and } p \nmid d,\ p\text{: odd} \\ \mathbb{F}_{p^2} & \text{if } d \notin (\mathbb{F}_p^\times)^2 \text{ and } p \nmid d,\ p\text{: odd} \Rightarrow p\mathcal{O}_K\text{: maximal} \\ \mathbb{F}_p[x]/x^2 & \text{if } p \mid d \\ \mathbb{F}_p[x]/((x-1)^2) & \text{if } p \nmid d,\ p=2 \end{cases}$$

For each case, it corr with:

- $\mathfrak{a}$ has two possibilities: $(p, 1+\sqrt{d})$, $(p, 1-\sqrt{d})$
- $\mathfrak{a} = p\mathcal{O}_K \Rightarrow [\mathcal{O}_K/\mathfrak{a} : \mathbb{F}_p] = 2$
- $(p, \sqrt{d})$
- $(2, \sqrt{d}-1)$

DEFINITION

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Let R be a commutative ring with identity, and let M, N be R-modules. We define the tensor product of M and N over R as:

$$M \otimes_R N = M \times N / \sim, \quad \begin{cases} (rm, n) \sim (m, rn) \\ (m+m', n) \sim (m,n) + (m', n) \\ (m, n+n') \sim (m,n) + (m, n') \end{cases}$$

It satisfies the following universal property: $\forall$ R-bilinear map $\varphi: M \times N \to L$, $\exists$ a unique R-module homo $\bar{\varphi}: M \otimes_R N \to L$, s.t.

$$\begin{tikzcd} M \times N \arrow[rr, "\varphi"] \arrow[dr] & \arrow[d, phantom, "\circlearrowleft"] & L \\ & M \otimes_R N \arrow[ur, "\bar{\varphi}"'] & \end{tikzcd}$$

DISCRIMINANT AND INTEGRAL BASES

Say $K = \mathbb{Q}(\theta)$ for some $\theta \in K^\times$, $\mathcal{O}_K$: free $\mathbb{Z}$-module of rank $[K:\mathbb{Q}]$

GOAL

1. Find a $\mathbb{Z}$-basis of $\mathcal{O}_K$
2. $\exists$? a basis of $\mathcal{O}_K$, s.t. $\mathcal{O}_K = \mathbb{Z}[\theta]$, $\theta \in \mathcal{O}_K$